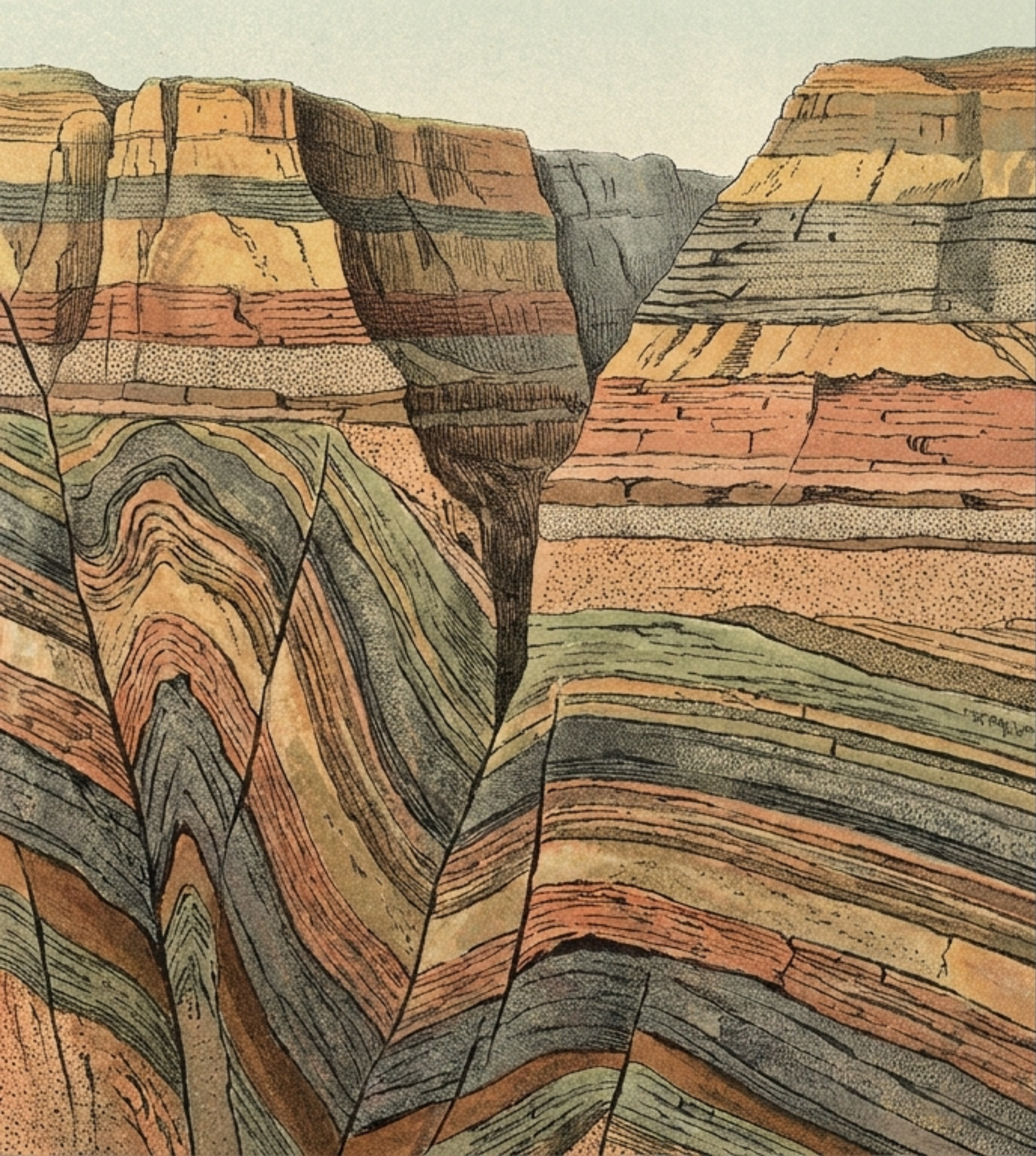




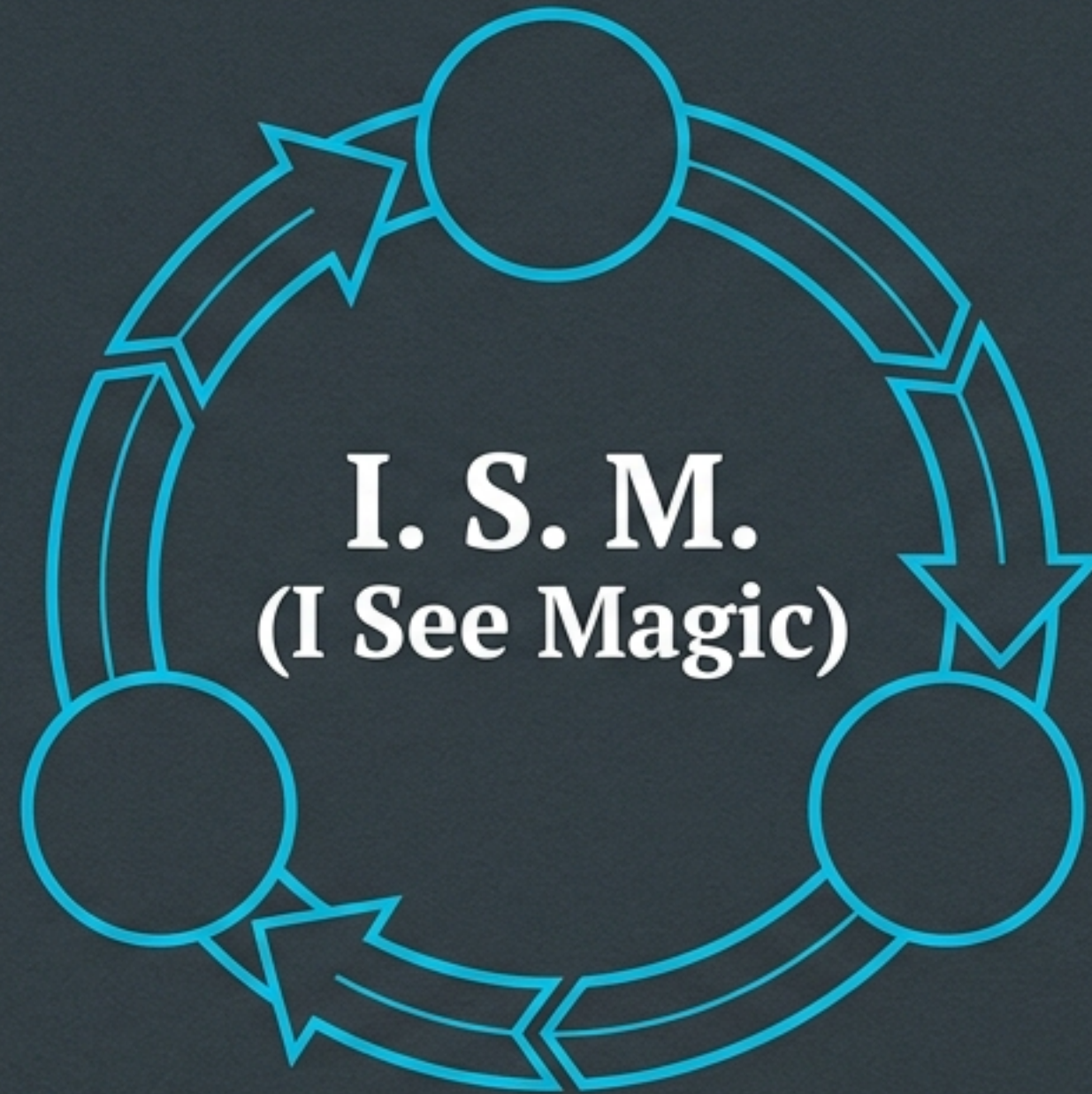
WEATHERING & SOIL FORMATION

*Geography Master Study Deck
(Active Recall Flashcards)*

HOW TO USE THIS DECK: Look at the question and the memory clue on the dark slides. Try to recite the exact answer aloud. Proceed to the next light slide to check your accuracy.

Question 1.a.

What is rock cycle? Explain.

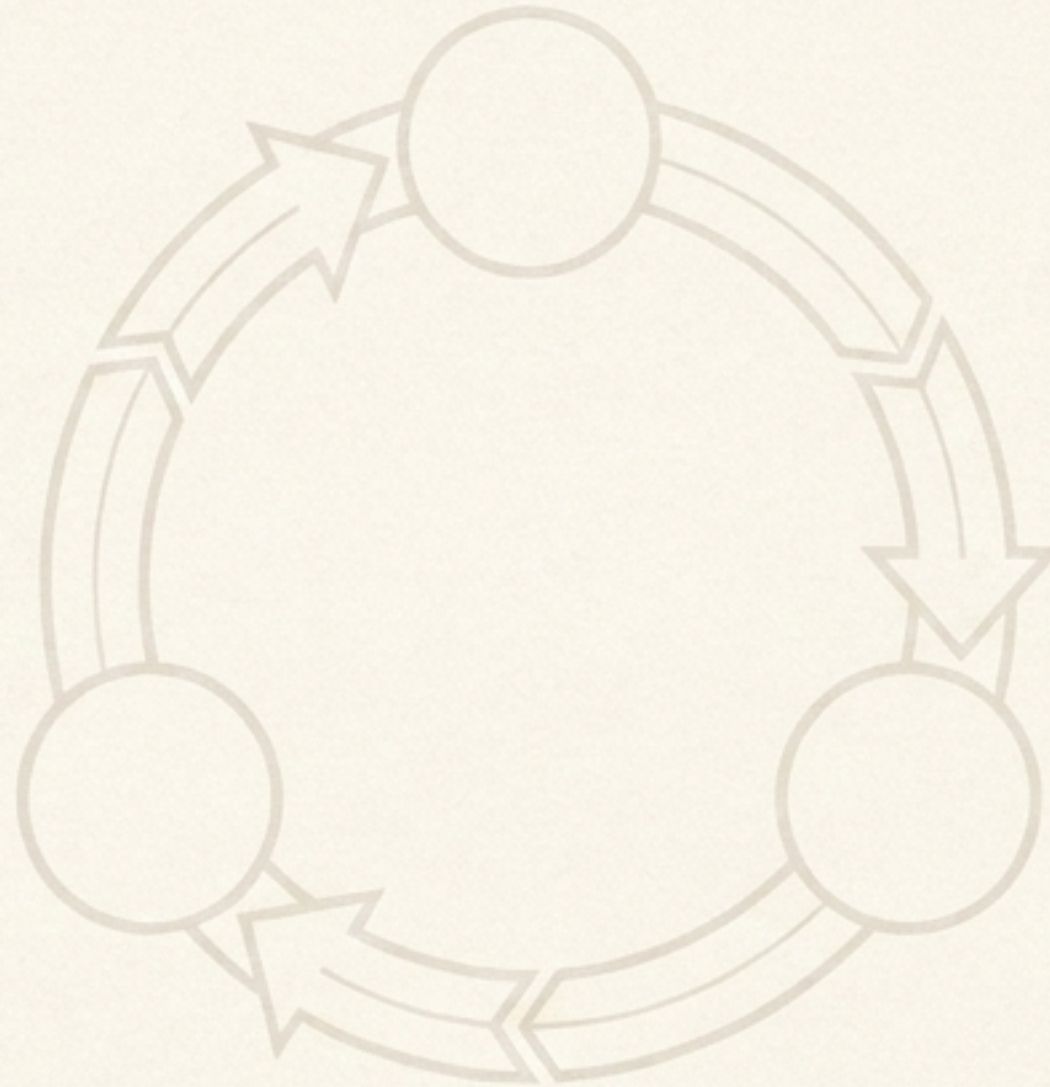


Remember the continuous circuit of **I. S. M.** —
Igneous, **Sedimentary**, and **Metamorphic** rocks.

Answer 1.a.

The three categories of rocks are constantly forming from one another in a **continuous circuit** which is known as **Rock cycle**.

When **igneous** rocks are exposed to weather changes, they are broken and eroded and give birth to **sedimentary** rocks.

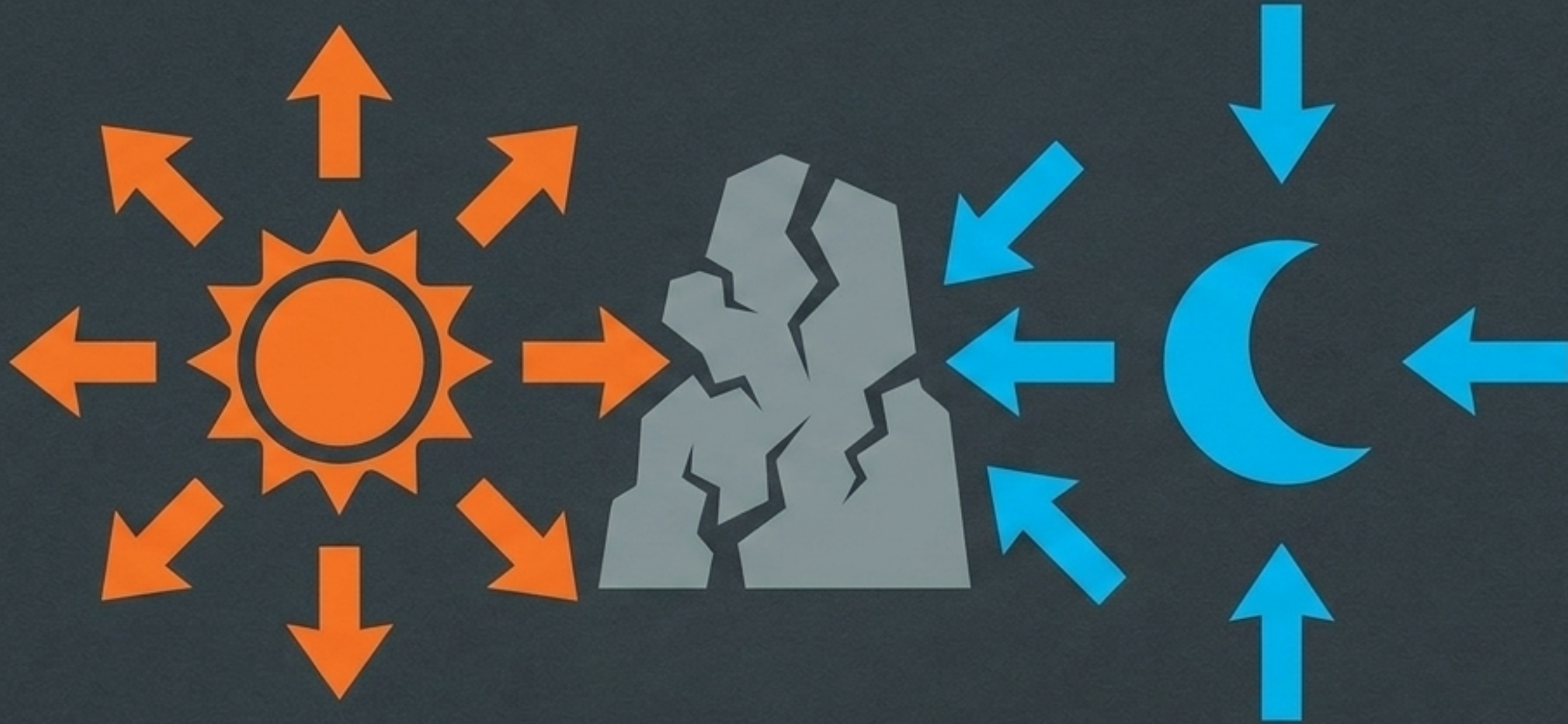


Due to excessive heat and pressure, these rocks further undergo a change into **metamorphic** rocks.

Thus, this cycle of change from one type of rock into another is called **Rock cycle**.

Question 1.b.

How does the change in temperature help in the process of weathering?



Think H.E.C.C. — High Expands, Cool Contracts.

Answer 1.b.

The rocks are composed of minerals which have different rates of heating and cooling.

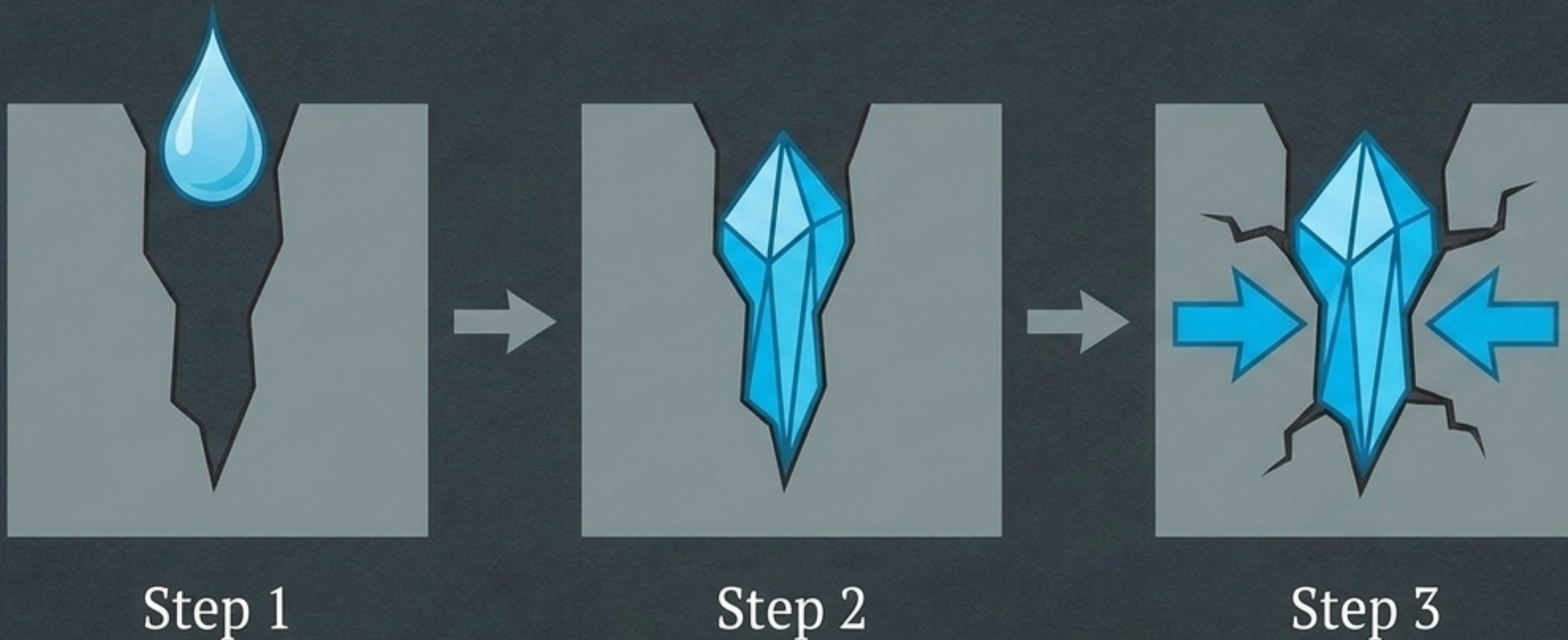
During day, the temperatures are **high and expansion** of rocks takes place.

During the night, the temperatures **fall and contraction** of the rocks takes place.

As the rate of expansion and contraction of the minerals of which the rock is composed, is different at various temperatures, the rocks break up into blocks or as grains depending on the rock structure.

Question 1.c.

Explain the process of weathering by frost action.



Remember the phrase: **W.F.E.C.** (Water Freezes, Expands, Cracks).

Answer 1.c.



Sometimes water gets collected in the cracks in the rocks.

When the temperature is low, this water freezes and expands and cracks the rocks but when the temperature is high, the volume of water is reduced due to melting of ice.

Freezing water, has increased volume that exerts pressure on the rock and widens the cracks... Continuous action causes the disintegration of the rocks.

Question 1.d.

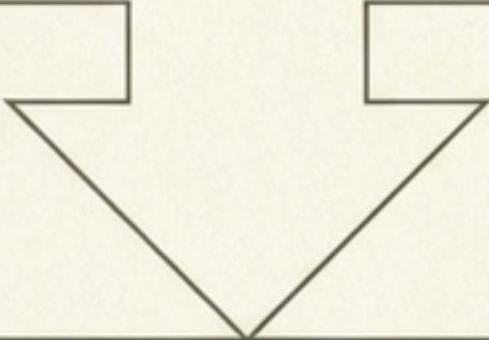
How does the chemical weathering effect the rocks?



Think **D.D.** - **P.C.** (Decay & Decomposition changes Physical & Chemical).

Answer 1.d.

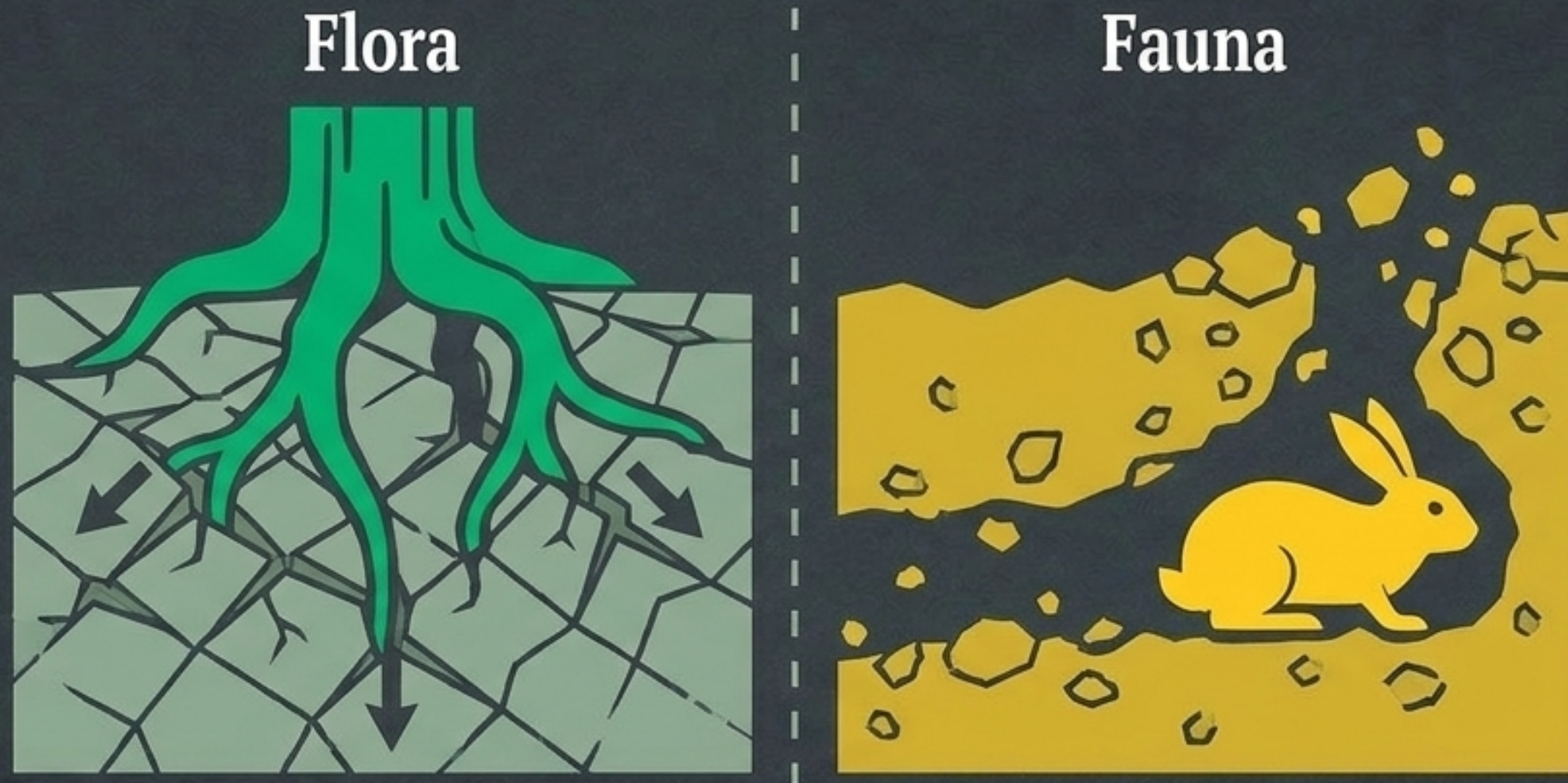
Chemical weathering results in the decay and decomposition of the rocks



...which involve a change in both the physical as well as chemical properties of the rock.

Question 1.e.

Mention the various ways in which plants and animals help in the process of weathering.



Memory Clue: **Roots Reach**, **Animals Alter** (Roots penetrate/widen, burrowing animals dig).

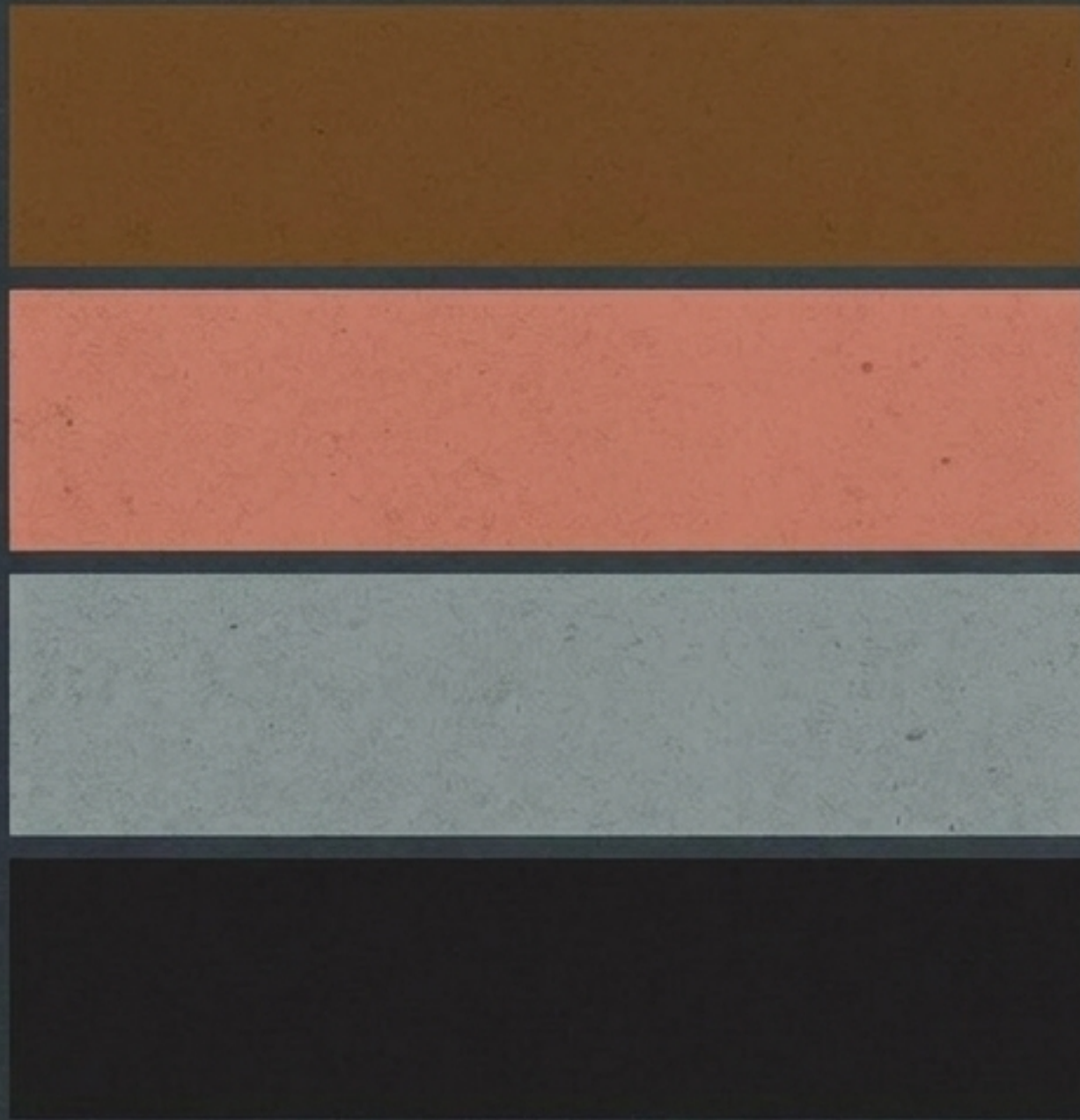
Answer 1.e.

By plants- “The **roots of the plants** and trees penetrate deep below the earth’s surface. They sometimes reach the cracks in the rocks and **widen the cracks** leading to physical disintegration of the rocks.”

By animals- “**Burrowing animals dig holes** or burrows and tunnels below the earth’s surface and this results in the gradual disintegration of the rocks.”

Question 1.f.

Describe a typical soil horizon.



Tiny Spiders Play Below

(Topsoil,

Subsoil,

Parent rock, Bedrock)

Answer 1.f.

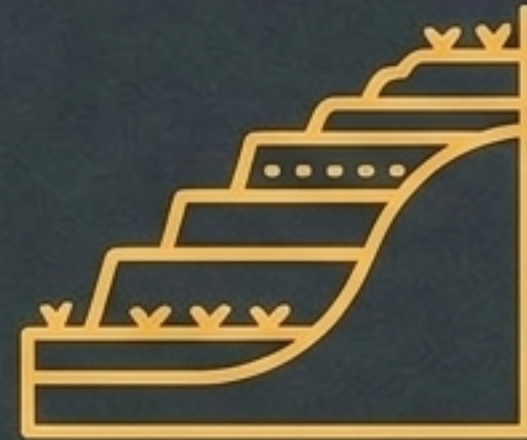
The vertical section of the soil showing different layers of it is known as soil profile.

There are different horizons of soil such as

- topsoil
- subsoil
- parent rock
- and bedrock.

Question 1.g.

List some methods of soil conservation.



A C R T (Action Conserves Rock & Timber) – Afforestation,
Checking overgrazing, Rotation, Terrace farming.

Answer 1.g.

Some of the measures to prevent the soil from erosion and fertility exhaustion are given alongside.

- Growing more trees on a large scale (**A**fforestation).
- **C**hecking of overgrazing.
- **R**otation of crops.
- Following the practice of **t**errace farming.